

High-Power Kerr Comb Source for Data Communications

Swarnava Sanyal^{1,*}, Michael Cullen², Songli Wang², Yuyang Wang^{2,3}, Yoshitomo Okawachi^{1,†},
Karl J. McNulty², Michal Lipson^{1,2}, Keren Bergman², and Alexander L. Gaeta^{1,2}

¹Department of Applied Physics and Applied Mathematics, Columbia University, New York, New York 10027, USA

²Department of Electrical Engineering, Columbia University, New York, New York 10027, USA

³Now with Dept. of Electrical and Computer Engineering, University of Connecticut, Storrs, CT 06269, USA

[†] Now at Xscape Photonics Inc.

Author e-mail address: s.sanyal@columbia.edu

Abstract: We demonstrate a high-power Kerr comb source with configurable 300-, 200-, and 100-GHz spacings. Direct 32 Gbps modulation of multiple channels yields open optical eyes, highlighting its potential for scalable, high-bandwidth photonic interconnects based on wavelength-division multiplexing.
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Microresonator-based Kerr frequency combs, with line spacings between 50 GHz to 1 THz offer the prospect of a multi-wavelength source driven by a single pump laser for wavelength-division multiplexing (WDM) data transmission that can replace an array of laser sources, thereby achieving massive parallelism over a small footprint [1,2]. Such combs realized in a CMOS-compatible platform can be further integrated with on-chip MZI filters and modulators to form a fully integrated silicon-photonics transmitter [3]. In recent years, Kerr combs operating in the normal group-velocity dispersion (GVD) regime have emerged as a promising solution for data transmission since they have been experimentally shown to have high pump-to-comb conversion efficiencies > 40%, high power-per-line, and tailorable spectral envelopes [4-6].

In this work, we demonstrate an efficient and high-power normal-GVD Kerr comb source with adjustable channel spacing and high spectral power uniformity. The high optical power-per-line for the comb source allows us to demonstrate direct comb line modulation at 32 Gbps with a commercial transmitter for multiple channels without the need for any additional optical amplification stage.

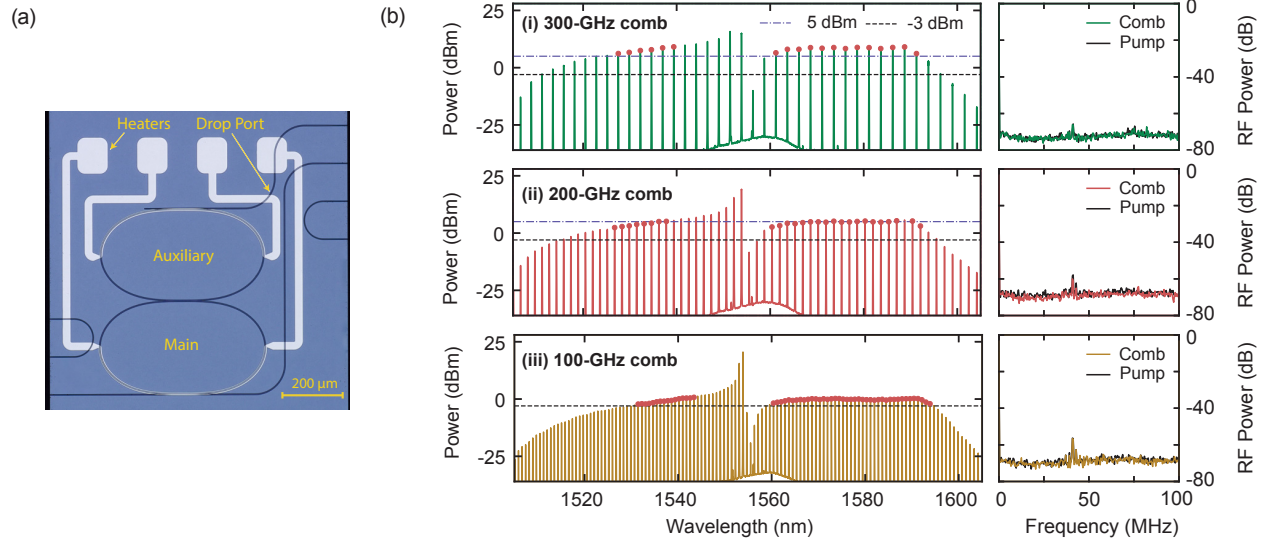


Fig. 1. (a) Microscope image of our SiN coupled-resonator-based normal-GVD comb chip. Integrated resistive heaters allows us to tune the resonance positions of the individual resonators. A drop port on the auxiliary resonator taps off excess power buildup caused by strong inter-resonator coupling. (b) Experimentally measured output optical and RF spectra for (i) 300-GHz, (ii) 200-GHz, and (iii) 100-GHz comb states with line markers at -3 dBm and 5 dBm, where the power levels correspond to the optical power in the bus waveguide. The red-filled circles correspond to the comb channels within 3dB power range. All the comb states exhibit low-noise comparable to that of the pump

1. Kerr Comb Source with Adjustable Channel Spacing

Figure 1(a) shows the microscope image of our silicon-nitride (SiN) coupled-resonator device, where the interaction between the modes in the main and auxiliary resonators results in mode splitting induced by avoided mode crossings (AMCs). This allows for the generation of initial parametric sidebands that initiate the comb formation in the normal-GVD regime. The main and auxiliary free-spectral-ranges (FSRs) are 100 GHz and 101.42 GHz, respectively, which corresponds to an AMC period of 53 nm. The SiN waveguide cross-section has 620-nm height and 1600-nm width, corresponding to a GVD of $46.2 \text{ ps}^2/\text{km}$ at the pump wavelength of 1553.8 nm. The resonator incorporates 90° adiabatic Euler bends to suppress coupling to higher-order modes that are supported in such relatively wide waveguide widths. The coupling gaps between the bus waveguide and main resonator, and between the main and auxiliary resonators, are 500 nm and 450 nm, respectively. The long interaction length allows for operation in the strongly overcoupled and strong mode-interaction regime, which enables us to achieve high pump-to-comb conversion efficiencies and a larger comb bandwidth [7], respectively. An auxiliary drop port is implemented to mitigate excess power buildup in the auxiliary resonator resulting from strong mode-coupling and ensure access to stable mode-locked comb states [7].

The SiN comb chip is pumped using a single-frequency continuous wave laser that is coupled into the chip using a lensed fiber. The main and auxiliary resonance locations can be adjusted using the integrated resistive heaters, which allows for tuning the AMC position relative to the fixed pump wavelength and deterministically control the channel spacing of the comb state [7,8]. A higher channel spacing is obtained when the AMC is located further away from the pump to the higher wavelength side. Figure 1b shows the measured optical spectra for (i) 300-GHz, (ii) 200-GHz and (iii) 100-GHz comb states generated using the same pump power of 375 mW. The power levels have been calibrated to correspond to the optical power in the bus waveguide. The corresponding RF-noise spectra show that all the comb states have a noise level comparable to that of the pump. Table 1 summarizes the key specifications of our comb source for the three channel spacings, including pump-to-comb conversion efficiency, the number of comb lines above specified power thresholds (5 dBm and -3 dBm), and spectral power uniformity. The latter is quantified by the number of channels within a 3 dB power range, which are highlighted in Fig. 1b with red-filled circles.

Comb State	Conv. Eff.	#Lines > 5dBm	#Lines > -3 dBm	#Lines within 3 dB
300 GHz	63.6%	25	36	19
200 GHz	48.1%	17	48	27
100 GHz	33.6%	-	73	57

Table 1. Comparison of the conversion efficiency, number of channels above 5dBm and -3 dBm, respectively and number of lines within 3dB of optical power for the 300-GHz, 200-GHz and 100-GHz comb state.

We observe a clear trade-off between channel spacing and comb power that has direct implications for photonic link design. For a given coupled-resonator design, larger comb spacings, such as 300 GHz, provide higher power-per-line and greater conversion efficiency, while smaller spacings, like 100 GHz, yield more channels but at lower comb-line powers. This trade-off implies that denser combs may require more sophisticated receiver architectures and advanced digital signal processing to handle the reduced channel power and signal-to-noise ratio. The ability to tune the channel spacing thus offers flexibility in designing high-bandwidth, energy-efficient DWDM links, allowing the optimal balance between channel count and per-channel power to be chosen based on transmitter/receiver design and link budget [9].

2. Direct Comb Line Modulation

We demonstrate direct modulation of multiple comb channels of the 300-GHz, 200-GHz and 100-GHz combs. Due to high optical power-per-line, no additional amplification stage was required before the transmitter, thus highlighting the efficiency of our comb source to be implemented in a DWDM link. A fiber add/drop filter centered at 1550 nm with a bandwidth of ± 6 nm was used to filter out the pump and nearby comb lines, and the remaining comb was sent to a commercial linear transmitter (Thorlabs MX35E). The modulation was performed by an Anritsu MP1900A pulse pattern generator generating a PRBS31 signal at 32 Gbps. A tunable bandpass filter (EXFO XTA-50) was used to selectively filter an individual, modulated comb line, after which it was amplified with an optical preamplifier to

compensate for the filter loss (approximately 6 dB) and measured by Keysight N1092C sampling oscilloscope. Figure 2a shows the zoomed-in optical spectra for the different channel spacing combs, with three highlighted comb channels—Ch 1568.6 nm, Ch 1573.6 nm, and Ch 1578.6 nm. The 300-GHz comb lines have approximately 3.7 dB and 8.6 dB higher optical power than the 200-GHz and 100-GHz comb lines, respectively. Figures 2b, 2c and 2d present the eye diagrams at 32 Gbps for three highlighted comb channels corresponding to the 300-GHz, 200-GHz and 100-GHz comb states, respectively. Open optical eyes were observed for all channels.

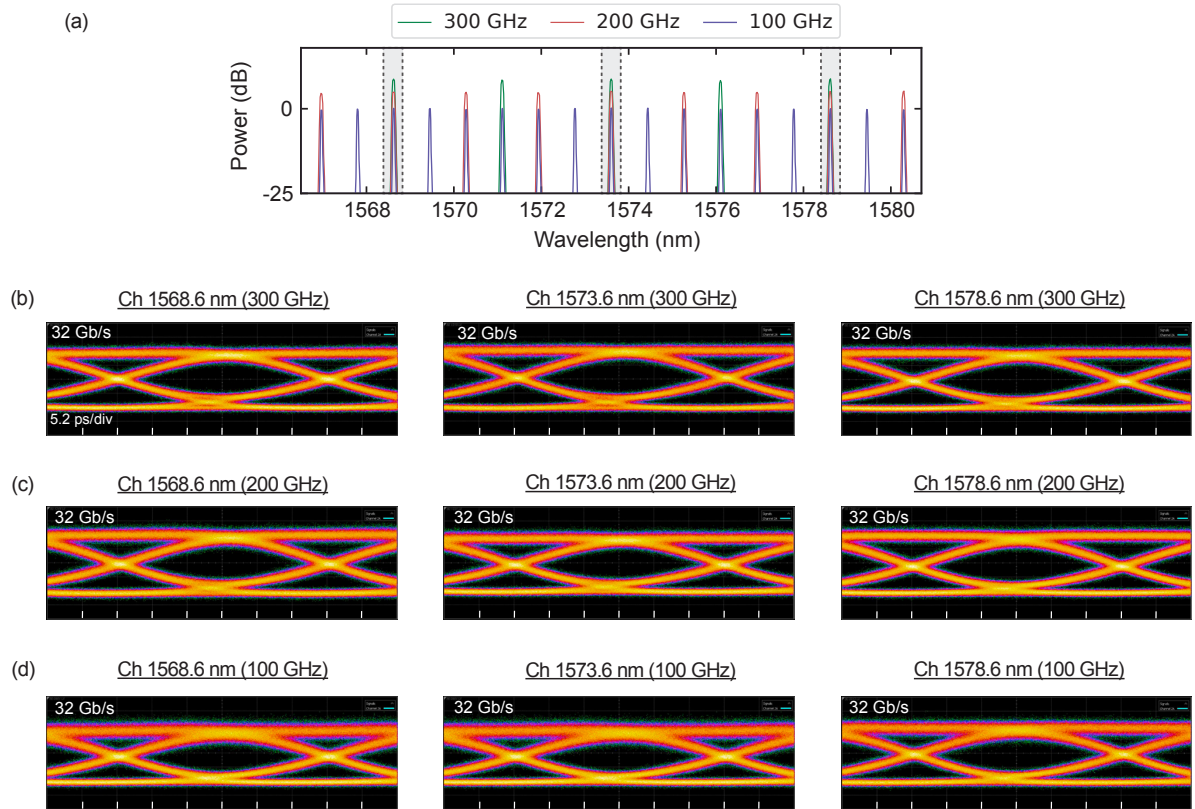


Fig. 2. (a) Zoomed-in optical spectra for the three comb states. (b)–(d) Eye diagrams measured at 32 Gbps using PRBS31, corresponding to direct modulation of the three highlighted channels in (a) for the 300-GHz, 200-GHz, and 100-GHz comb states, respectively.

In conclusion, we have demonstrated a high-power, low-noise, and efficient normal-GVD Kerr comb source with adjustable channel spacings of 300 GHz, 200 GHz, and 100 GHz. The 300-GHz comb achieved the highest efficiency (>60%) with 25 lines above 5 dBm, 19 of which lie within a 3 dB power range. In contrast, the 100-GHz comb exhibited a lower efficiency of 33.6% but produced 73 lines above -3 dBm, with 57 within 3 dB, highlighting a trade-off between channel spacing and comb-line power. We further demonstrated direct 32 Gbps modulation of three comb channels for each comb state, showing open optical eyes. The ability to deterministically tune the channel spacing of our comb source underscores its flexibility for integration into a wide range of photonic links for hyperscale computing systems.

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